



Step-by-Step: Optimize the data-cleaning process

This reading outlines steps the instructor performs in the following video, [Optimize the data-cleaning process](#). The video teaches some useful spreadsheet functions, which can make your data-cleaning even more successful.

Keep this step-by-step guide open as you watch the video. It can serve as a helpful reference if you need additional context or clarification while following the video steps. This is not a graded activity, but you can complete these steps to practice the skills demonstrated in the video.

What you'll need

If you would like to access the spreadsheet the instructor uses in this video, click the link to the dataset to create a copy. If you don't have a Google account, you may download the data directly from the attachments below.

Link to logistics data: [International Logistics Association Memberships - Data for Cleaning](#)

Link to cosmetics data: [Cosmetics Inc. - Data for Cleaning](#)

OR

Example 1: The COUNTIF function

COUNTIF is a spreadsheet function that returns the number of cells within a range that match a specified value.

Use COUNTIF to find numbers lower than 100

1. Open the International [Logistics Association Memberships - Data for Cleaning](#) dataset, and scroll down to row 74.
 - a. Note: The dataset has 72 rows, and row 73 is left blank for separation.
2. In cell H74, enter Member Dues < 100 to label the calculation.
3. In cell I74, enter the formula `=COUNTIF(I2:I72, "<100")` to count how many members in the cell range I2:I72 pay dues of less than \$100. This formula returns a value of 1, indicating one value is below \$100.
4. In cell I55, change -\$200 to \$200. Cell I74 should now display the value 0.

Use COUNTIF to find numbers higher than 500

1. In cell H75, enter Member Dues > 500.
2. In cell I75, enter the formula `=COUNTIF(I2:I72, ">500")` to count how many members in cell range I2:I72 pay dues of greater than 500. This formula returns a value of 1, indicating one value is above 500.
3. In cell I44, change \$1,000 to \$100. Cell I75 should now display the value 0.

Example 2: The LEN function

The LEN function is useful if you have a certain piece of information in your spreadsheet that you know must contain a certain length.

1. Right click cell A.

2. Select + Insert one column right to create a new, empty column.
3. Select cell B1 and enter LEN to name the new column.
4. In cell B2, enter `=LEN(A2)`. This function references the value of cell A2 and returns its length, 6.
5. Double-click on the lower right corner of cell B2. This will copy the function through the rest of the column. Each cell will show the length of the Member ID in that row.

Example 3: Use conditional formatting

Conditional formatting is a spreadsheet tool that changes how cells appear when values meet specific conditions.

1. To highlight all of column B except for the header, select cell B. Then press CONTROL (Windows) or COMMAND (MAC) and select cell B1.
2. Navigate to the Format menu, and choose Conditional Formatting.
3. Set the Format rules field to Is not equal to and enter 6 as the value.
4. Select Done.
5. Notice cell B36 is highlighted because its value is 7.

Example 4: The LEFT and RIGHT functions

LEFT is a function that returns a set number of characters from the left side of a text string. **RIGHT** is a function that returns a set number of characters from the right side of a text string.

The LEFT function

1. Use the [Cosmetics Inc. - Data for Cleaning](#) dataset.
2. Select cell H1, and enter Left.
3. In cell H2, enter `=LEFT(A2, 5)` to extract the first five characters from cell A2. This function will show the substring 51993.

4. Select cell H2.
5. Select and hold the fill handle, the small circle in the corner of a selected cell, then drag this formula down to populate the rest of this column.

The **RIGHT** function

1. Select cell I1, and enter Right.
2. In cell I2, enter **=RIGHT (A2 , 4)** to extract the last four characters from cell A2.
This function will show the substring Masc.
3. Select cell I2.
4. Select and hold the fill handle and drag this formula down to populate the rest of this column.

Example 5: The **MID** function

MID is a function that returns a segment from the middle of a text string.

1. Select cell J1, and enter Mid.
2. In cell J2, enter **=MID (D2 , 4 , 2)** to extract the two-letter state code that starts at character four in cell D2.
3. Double-click the fill handle and to automatically populate the rest of this column.

Example 6: The **CONCATENATE** function

CONCATENATE is a spreadsheet function that joins together two or more text strings.

1. Select cell K1, and enter Concatenate.
2. In cell K2, enter **=CONCATENATE (H2 , I2)** to combine the values from columns H and I.
3. Double-click the fill handle and to automatically populate the rest of this column.

Example 7: TRIM function

TRIM is a function that removes leading, trailing, and repeated spaces in data.

1. Select cell L1, and enter Trim.
2. In cell L2, enter **=TRIM(C2)** to remove any leading, trailing, or repeated spaces.
3. Double-click the fill handle and to automatically populate the rest of this column.

Key Concepts:

- **LEN:** The LEN function returns the length of a text string, which is useful for validating data formats.
- **CONCATENATE:** The CONCATENATE function joins two or more text strings into one string.
- **COUNTIF:** The COUNTIF function counts the number of cells within a range that meet a specified condition.
- **MID:** The MID function returns a segment from the middle of a text string based on specified starting position and length.
- **LEFT:** The LEFT function extracts a specified number of characters from the beginning of a text string.
- **TRIM:** The TRIM function removes leading, trailing, and repeated spaces from text, ensuring clean data entries.
- **RIGHT:** The RIGHT function extracts a specified number of characters from the end of a text string.